

Task: 12 - Stimulating gaming concepts in Python

Aim:

To stimulate gaming concepts using pygame

Snake game:

Problem 1: Program to create a snake game using Pygame

Conditions:

1. Set the window size
2. Create a snake and make the snake to move in ^{all} the directions. When the snake hits fruit, increase the score by 1.
3. If the snake hits the window. Game over

Algorithm:

1. Import package pygame and initialize it. Define window size
2. Create a snake class which initializes the snake position, colour and movement. Also create a fruit class which initializes the fruit position and colour
3. Create a function to check if the snake collides with the fruit and increase the score
4. Create a function to check if the snake collides with the window and end the game
5. Create a function to update the snake position and to game display to draw the snake & fruit. Create a loop to update it and check for collisions
6. End the game if the user quits or if the snake collides with the window.

Program:

```
import pygame
import time
import random
snake_speed = 15
window_x = 720
```

- (14/8/20)

3. Add labels and set the size of text boxes
 4. Create a button and get the values of
 the user for you when submitting the values
 5. Get it and close the main window when the
 button is clicked

Output:

```

import tkinter as tk
root = tk.Tk()
root.title("Text Box Input")
label1 = tk.Label(root, text="Enter value 1")
entry1 = tk.Entry(root)
label2 = tk.Label(root, text="Enter value 2")
entry2 = tk.Entry(root)
label3 = tk.Label(root, text="Enter value 3")
entry3 = tk.Entry(root)
entry1.grid(row=1, column=1)
entry2.grid(row=1, column=2)
entry3.grid(row=1, column=3)

```

def get_values():
 val1 = entry1.get()
 val2 = entry2.get()
 val3 = entry3.get()
 print("Value 1:", val1)
 print("Value 2:", val2)
 print("Value 3:", val3)
 submit_button = tk.Button(root, text="Submit", command=get_values)

VEL TECH	
EX NO.	11
PERFORMANCE (%)	2
RESULT AND MARKS (%)	2
VIVA VOCE (%)	2
RECORD (%)	2
TOTAL (%)	10
SIGN WITH DATE	

label1.pack() entry1.pack()
 label2.pack() entry2.pack()
 label3.pack() entry3.pack()
 submit_button.pack()

Result: This program is for
 to get the values of the user


```

black = pygame.Color(0,0,0)
white = pygame.Color(255,255,255)
red = pygame.Color(255,0,0)
green = pygame.Color(0,255,0)
blue = pygame.Color(0,0,255)
pygame.init()
pygame.display.set_caption('neekforhelk's snakes')
game_window = pygame.display.set_mode((window_x, window_y))
fps = pygame.time.Clock()
snake_position = [100, 50]
snake_body = [[100, 50], [90, 50], [80, 50], [70, 50]]
fruit_position = [random.randrange(1, (window_x//10))*10, random.randrange(1, (window_y//10))*10]
fruit_spawn = True
direction = 'RIGHT'
change_to = direction
score = 0

def show_score(choice, color, font, size):
    score_font = pygame.font.SysFont(font, size)
    score_surface = score_font.render("Score: " + str(score), True, color)
    score_rect = score_surface.get_rect()
    game_window.blit(score_surface, score_rect)

def game_over():
    myfont = pygame.font.SysFont('timesnewroman', 50)
    game_over_surface = myfont.render('Your score is: ' + str(score), True, red)
    game_over_rect = game_over_surface.get_rect()
    game_over_rect.midtop = (window_x/2, window_y/4)
    time.sleep(2)
    pygame.quit()

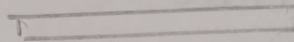
while True:
    for event in pygame.event.get():
        if event.type == pygame.KEYDOWN:
            if event.key == pygame.K_UP:
                change_to = 'UP'
            if event.key == pygame.K_DOWN:
                change_to = 'DOWN'
            if event.key == pygame.K_LEFT:
                change_to = 'LEFT'
            if event.key == pygame.K_RIGHT:
                change_to = 'RIGHT'

    if direction == 'UP':
        snake_position[1] -= 10

```

Output:

Score: 0




```

snake_position = 10
if direction == 'LEFT':
    snake_position -= 10
if direction == 'RIGHT':
    snake_position += 10
snake_body.insert(0, list(snake_position))
if snake_position[0] == fruit_position[0] and snake_position[1] == fruit_position[1]:
    score += 10
    fruit_spawn = False
else:
    snake_body.pop()
if not fruit_spawn:
    fruit_position = (random.randrange(1, (window_x//10)) * 10, random.randrange(1, (window_y//10)) * 10)
    fruit_spawn = True
    game_window.fill('black')
    for pos in snake_body:
        pygame.draw.rect(game_window, green, pygame.Rect(pos[0], pos[1], 10, 10))
    pygame.draw.rect(game_window, white, pygame.Rect(fruit_position[0], fruit_position[1], 10, 10))
    if snake_position[0] < 0 or snake_position[0] > window_x - 10:
        game_over()
    if snake_position[1] < 0 or snake_position[1] > window_y - 10:
        game_over()
    for block in snake_body[1:]:
        if snake_position[0] == block[0] and snake_position[1] == block[1]:
            game_over()
    show_score(1, white, 'times new roman', 20)
    pygame.display.update()
    fps.tick(snake_speed)

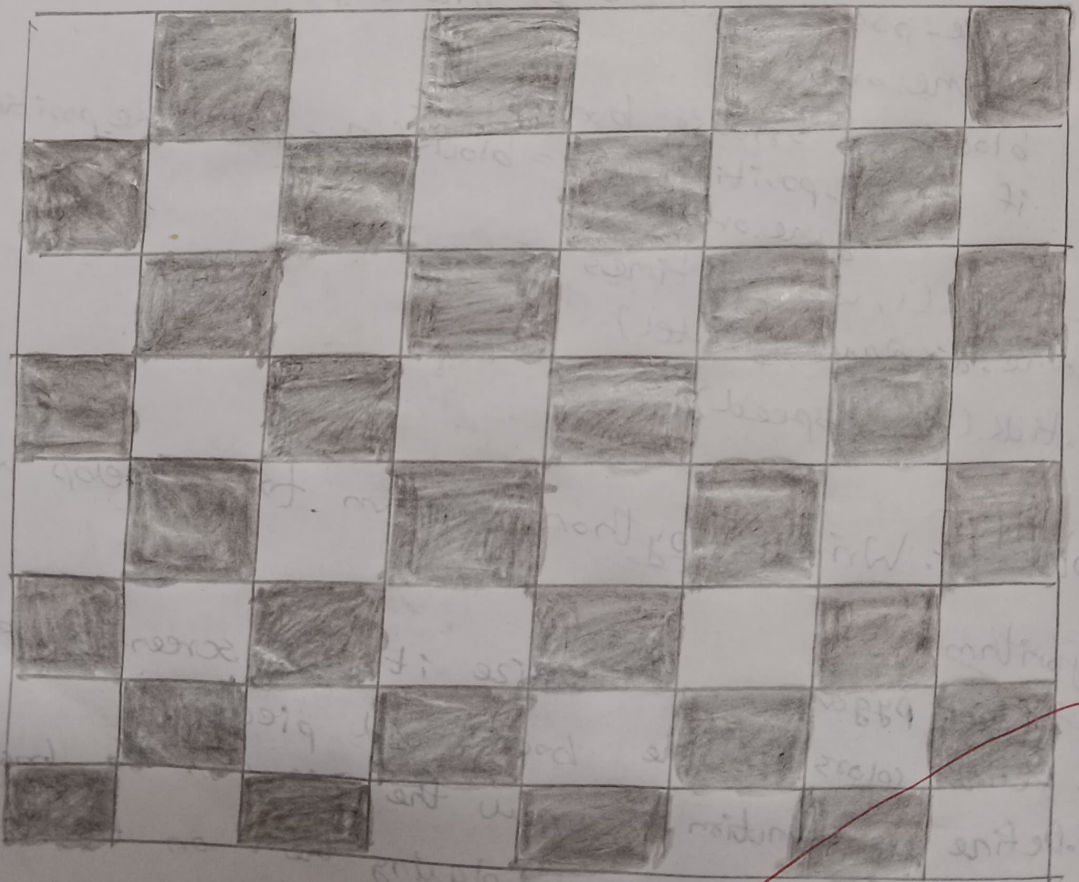
```

Problem 2: Write a python program to develop a chess board

Algorithm:

1. Import pygame and initialize it. Set screensize and title
2. Define colors for the board and pieces.
3. Define a function to draw the pieces on the board by loading images for each piece and placing them on the corresponding squares
4. Define the initial state of the board as a list of lists
5. Draw the board and pieces on the screen
6. Start the game loop

Output =



Program:

```
import pygame
pygame.init()
screen_size = (640, 640)
screen = pygame.display.set_mode(screen_size)
pygame.display.set_caption('Chess Board')
black = (0, 0, 0)
white = (255, 255, 255)
brown = (153, 76, 0)

def draw_board():
    for row in range(8):
        for col in range(8):
            square_color = white if (row + col) % 2 == 0 else brown
            square_rect = pygame.Rect(col * 80, row * 80, 80, 80)
            pygame.draw.rect(screen, square_color, square_rect)

def draw_pieces(board):
    piece_images = {'r': pygame.image.load('images/rook.png'), 'n': pygame.image.load('images/knight.png'), 'b': pygame.image.load('images/bishop.png'), 'q': pygame.image.load('images/queen.png'), 'k': pygame.image.load('images/king.png'), 'p': pygame.image.load('images/pawn.png')}
    for row in range(8):
        for col in range(8):
            piece = board[row][col]
            if piece != '':
                piece_image = piece_images[piece]
                piece_rect = pygame.Rect(col * 80, row * 80, 80, 80)
                screen.blit(piece_image, piece_rect)

board = [['r', 'n', 'b', 'q', 'k', 'b', 'n', 'r'], ['p', 'p', 'p', 'p', 'p', 'p', 'p', 'p'], ['', '', '', '', '', '', '', ''], ['', '', '', '', '', '', '', ''], ['', '', '', '', '', '', '', 'p'], ['p', 'p', 'p', 'p', 'p', 'p', 'p'], ['p', 'w', 'b', 'a', 'k', 'b', 'n', 'r']]

draw_board()
draw_pieces(board)

while True:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            pygame.quit()
            quit()
    pygame.display.update()
```

Result:

Thus the program is executed and verified successfully.

VEL TECH	
EX-NO	12
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	
TOTAL (20)	
SIGN WITH DATE	15